

Data Flow Diagrams & User Stories

Design Phase II

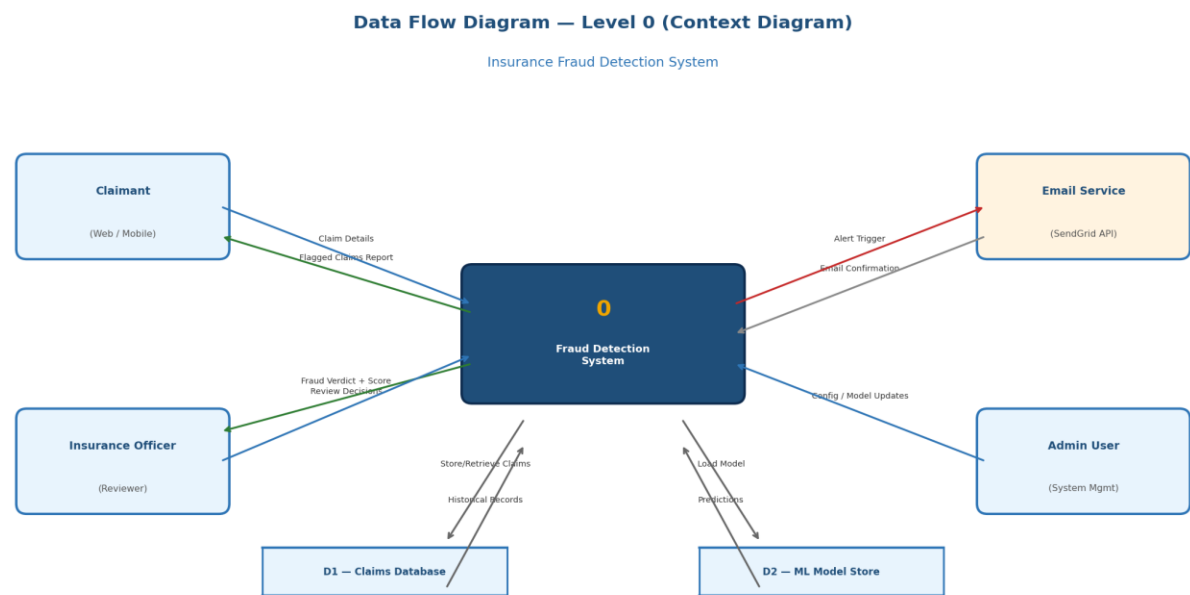
Date	07 March 2026
Team ID	
Project Name	Insurance Fraud Detection Using Machine Learning
Maximum Marks	4 Marks

Data Flow Diagrams

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD depicts the right amount of system requirements graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Level 0 — Context Diagram

Level 0 (Context Diagram) shows the entire Insurance Fraud Detection system as a single process, and shows the data flowing between it and the external entities (users, services, and databases).



DFD Level 0 — Entity & Data Flow Description

Entity	Type	Input to System	Output from System
Client (Web / Mobile User)	External Entity	Claim details (vehicle/health/property), personal information, supporting documents	Fraud probability score, fraud verdict (FRAUD/GENUINE), claim status, notifications
Insurance Officer (Reviewer)	External Entity	Review decisions (approve/reject/escalate), feedback on incorrect predictions	List of flagged high-risk claims, fraud analytics reports, daily summary

Entity	Type	Input to System	Output from System
ML Fraud Detection Engine	Internal Process	Pre-processed claim feature vectors from the data pipeline	Fraud probability score (0.0–1.0), risk label, top contributing features
Clients Database (D1)	Data Store	Historical claim records, user data, prediction logs	Training data for ML model, claim lookup for reviewer
ML Model Store (D2)	Data Store	Newly trained model artefacts and version metadata	Active model loaded for real-time inference
Email Service (SendGrid API)	External Entity	Fraud alert trigger from the system when score > threshold	Confirmation of email sent; fraud alert email delivered to officer
Admin User	External Entity	System configuration, user management commands, model update triggers	System health metrics, audit reports, user activity logs

DFD Level 1 — Process Decomposition

Level 1 breaks the central 'Fraud Detection System' process into its major sub-processes, showing how data flows between them internally.

Process No.	Process Name	Input Data Flows	Output Data Flows	Data Stores Used
P1	User Authentication	Login credentials, registration data	JWT access token, user session	D1 — Claims DB (users table)
P2	Claim Data Ingestion	Raw claim form fields, uploaded documents	Validated and structured claim record	D1 — Claims DB (claims table)
P3	Data Preprocessing	Raw claim record from P2	Normalised feature vector ready for ML model	None (in-memory pipeline)
P4	Fraud Prediction	Normalised feature vector from P3	Fraud probability score, risk label, SHAP feature importance	D2 — ML Model Store
P5	Result Storage & Audit	Fraud score + claim details from P4	Stored prediction record with timestamp and model version	D1 — Claims DB (predictions table)
P6	Notification Dispatch	Fraud score > threshold trigger from P4	Email alert dispatched via SendGrid API	None (stateless API call)
P7	Analytics & Reporting	Aggregated prediction records from D1	Dashboard charts, trend data, export files (PDF/CSV)	D1 — Claims DB

User Stories

User stories describe what each type of user needs to accomplish within the system. They follow the format: As a [user type], I can [action], so that [benefit].

User Type	Epic	USN	User Story / Task	Acceptance Criteria	Priority	Sprint
Claimant (Web User)	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	Can access my dashboard after successful registration.	High	Sprint-1

User Type	Epic	USN	User Story / Task	Acceptance Criteria	Priority	Sprint
Claimant	Registration	USN -2	As a user, I will receive a confirmation email once I have registered for the application.	Receive confirmation email and can click to verify account.	High	Sprint-1
Claimant	Registration	USN -3	As a user, I can register for the application through Google OAuth.	Can register and access dashboard using Google account.	Medium	Sprint-1
Claimant	Login	USN -4	As a user, I can log into the application by entering my email and password.	Redirected to dashboard on successful login. Error shown on failure.	High	Sprint-1
Claimant	Claim Submission	USN -5	As a user, I can submit an automobile insurance claim with vehicle and damage details.	Claim is saved and fraud probability score is displayed instantly.	High	Sprint-1
Claimant	Claim Submission	USN -6	As a user, I can submit a health insurance claim with treatment and billing details.	Health claim is processed by ML model and fraud verdict shown.	High	Sprint-2
Claimant	Claim Submission	USN -7	As a user, I can submit a property insurance claim with damage and valuation details.	Property claim fraud verdict and risk level displayed clearly.	High	Sprint-2
Claimant	Dashboard	USN -8	As a user, I can view all my previously submitted claims with their fraud scores and verdicts.	Claims listed in a table with score, verdict, date, and status.	Medium	Sprint-2
Claims Officer	Review	USN -9	As an officer, I can view all high-risk flagged claims assigned to me for review.	Flagged claims visible in my review queue with risk levels.	High	Sprint-2
Claims Officer	Review	USN -10	As an officer, I can approve, reject, or escalate a claim after reviewing the fraud score and explanation.	Decision is logged with timestamp and officer notes.	High	Sprint-2

User Type	Epic	USN	User Story / Task	Acceptance Criteria	Priority	Sprint
Claims Officer	Analytics	USN -11	As an officer, I can view fraud trend charts showing fraud rates by category and time period.	Charts display auto/health/property fraud rates with date filters.	Medium	Sprint-3
Admin	Management	USN -12	As an admin, I can create, update, and deactivate user accounts and assign roles.	User list visible; CRUD operations work; role changes take effect immediately.	Low	Sprint-3
Admin	Model Mgmt	USN -13	As an admin, I can trigger a model retraining job with the latest labelled claim data.	Retraining job runs; new model version deployed; accuracy metrics shown.	Low	Sprint-4
Claims Officer	Notifications	USN -14	As an officer, I receive an email alert when a claim scores above the fraud threshold.	Email received within 60 seconds of high-risk claim submission.	Medium	Sprint-4
Claimant (Web User)	Export	USN -15	As a user, I can export my claim history as a PDF or CSV file.	Download starts immediately; file contains all claims with scores.	Low	Sprint-4